

CS7015/CS6910 (Deep Learning) : Lecture 1

(Partial/Brief) History of Deep Learning

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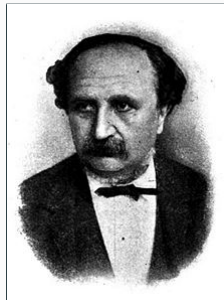
Acknowledgements

- Most of this material is based on the article “Deep Learning in Neural Networks: An Overview” by J. Schmidhuber^[1]
- **The errors, if any, are due to me and I apologize for them**
- Feel free to contact me if you think certain portions need to be corrected (please provide appropriate references)

Chapter 1: Biological Neurons

Reticular Theory

Joseph von Gerlach proposed that the nervous system is a single continuous network as opposed to a network of many discrete cells!



1871-1873

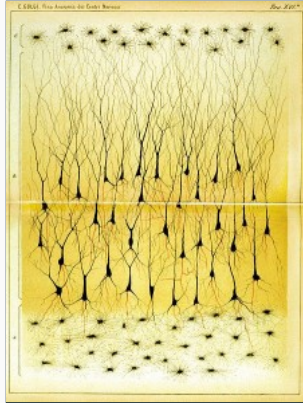


Reticular theory

Staining Technique

Camillo Golgi discovered a chemical reaction that allowed him to examine nervous tissue in much greater detail than ever before

He was a proponent of Reticular theory.



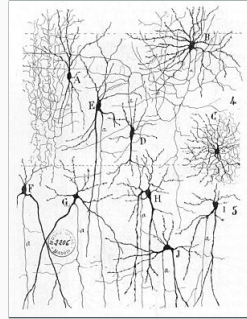
1871-1873



Reticular theory

Neuron Doctrine

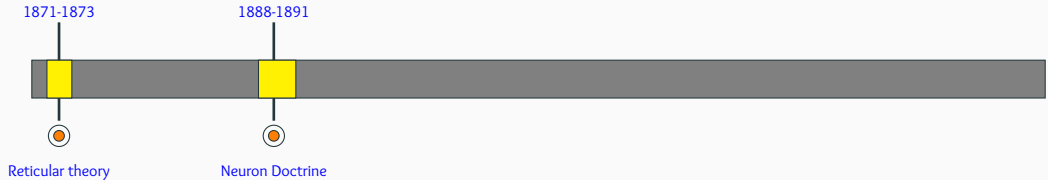
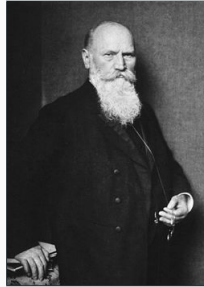
Santiago Ramón y Cajal used Golgi's technique to study the nervous system and proposed that it is actually made up of discrete individual cells forming a network (as opposed to a single continuous network)



The Term Neuron

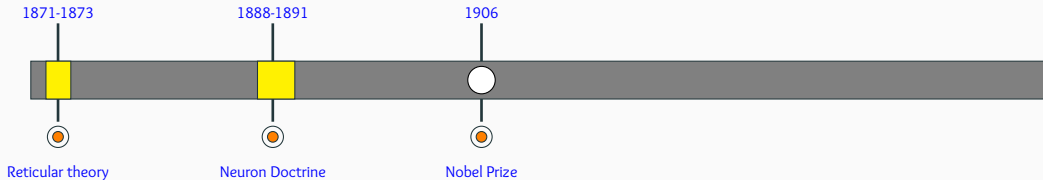
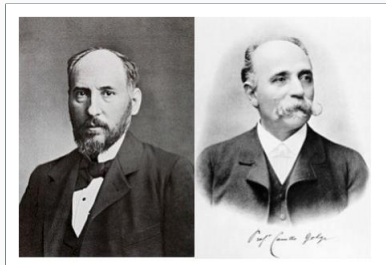
The term neuron was coined by Heinrich Wilhelm Gottfried von Waldeyer-Hartz around 1891.

He further consolidated the Neuron Doctrine.



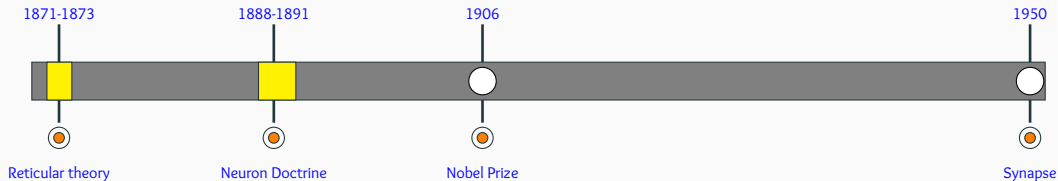
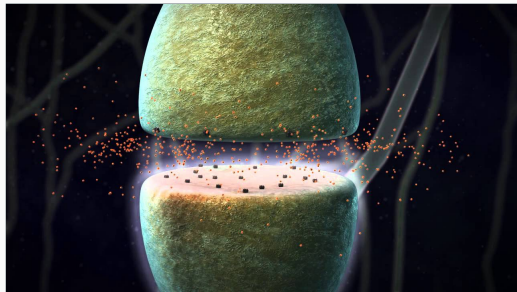
Nobel Prize

Both Golgi (reticular theory) and Cajal (neuron doctrine) were jointly awarded the 1906 Nobel Prize for Physiology or Medicine, that resulted in lasting conflicting ideas and controversies between the two scientists.



The Final Word

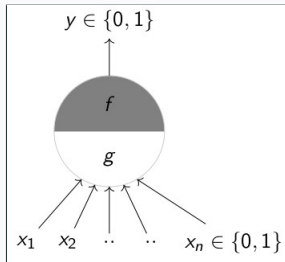
In 1950s electron microscopy finally confirmed the neuron doctrine by unambiguously demonstrating that nerve cells were individual cells interconnected through synapses (a network of many individual neurons).



Chapter 2: From Spring to Winter of AI

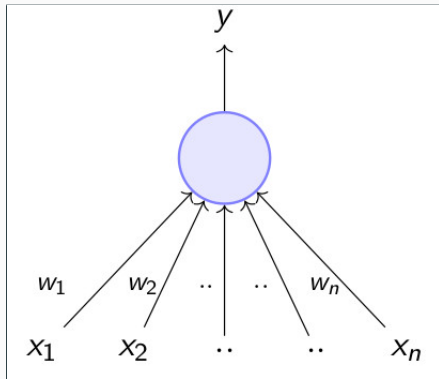
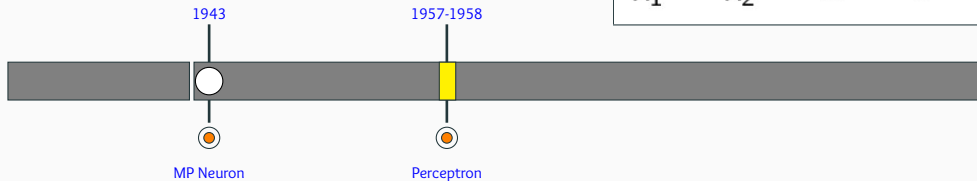
McCulloch Pitts Neuron

McCulloch (neuroscientist) and Pitts (logician) proposed a highly simplified model of the neuron (1943) ^[2]



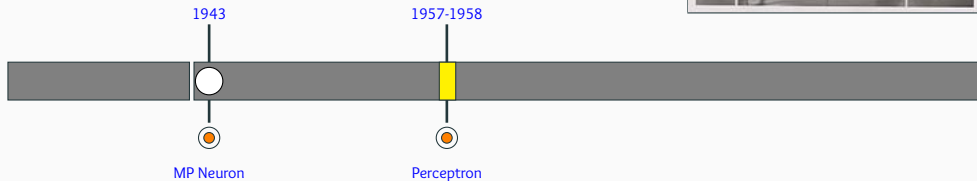
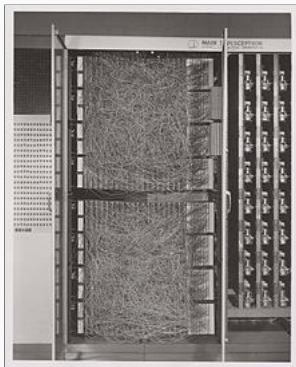
Perceptron

“the perceptron may eventually be able to learn, make decisions, and translate languages” -Frank Rosenblatt



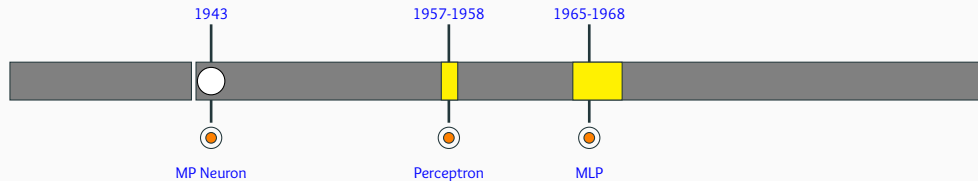
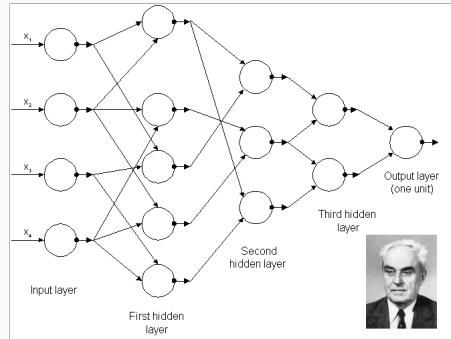
Perceptron

“the embryo of an electronic computer that the Navy expects will be able to walk, talk, see, write, reproduce itself and be conscious of its existence.” -New York Times



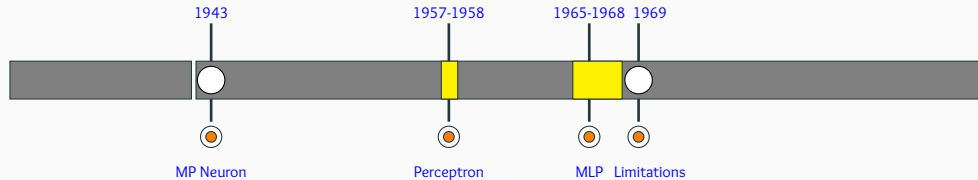
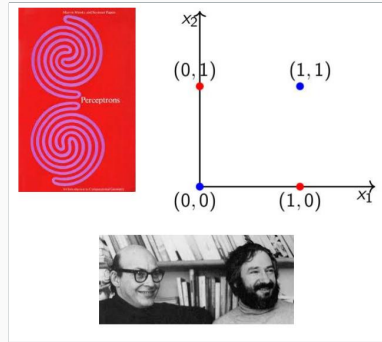
First generation Multilayer Perceptrons

Ivakhnenko et. al. [3]



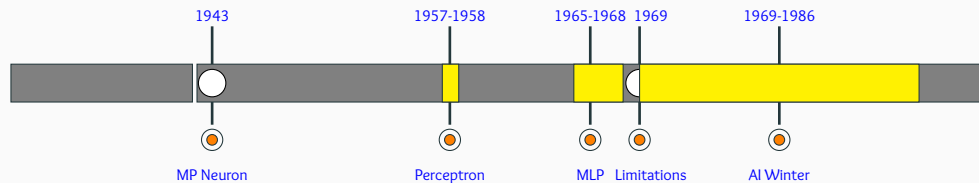
Perceptron Limitations

In their now famous book “Perceptrons”, Minsky and Papert outlined the limits of what perceptrons could do^[4]



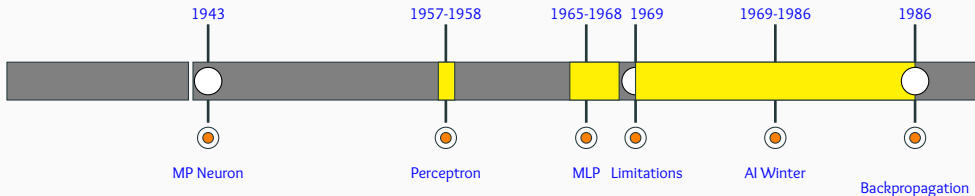
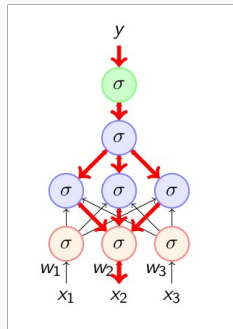
AI Winter of connectionism

Almost lead to the abandonment of
connectionist AI



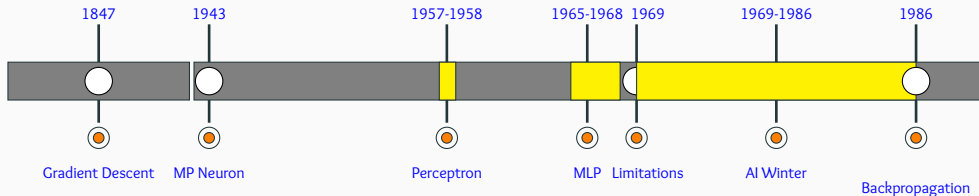
Backpropagation

- Discovered and rediscovered several times throughout 1960's and 1970's
- Werbos(1982)^[5] first used it in the context of artificial neural networks
- Eventually popularized by the work of Rumelhart et. al. in 1986^[6]



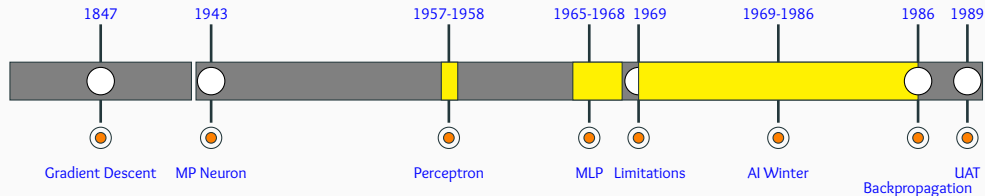
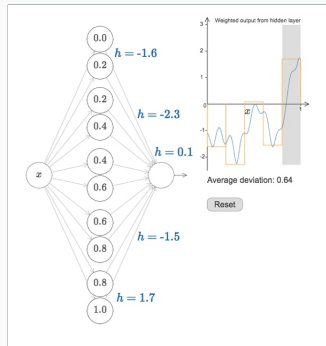
Gradient Descent

Cauchy discovered Gradient Descent
motivated by the need to compute the orbit
of heavenly bodies



Universal Approximation Theorem

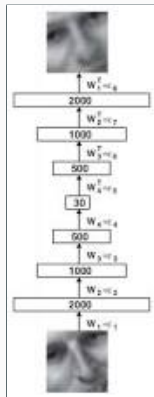
A multilayered network of neurons with a single hidden layer can be used to approximate any continuous function to any desired precision^[7]



Chapter 3: The Deep Revival

Unsupervised Pre-Training

Hinton and Salakhutdinov described an effective way of initializing the weights that allows deep autoencoder networks to learn a low-dimensional representation of data. [8]



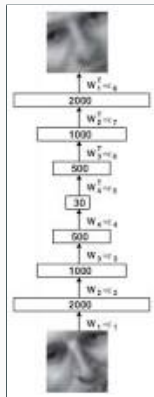
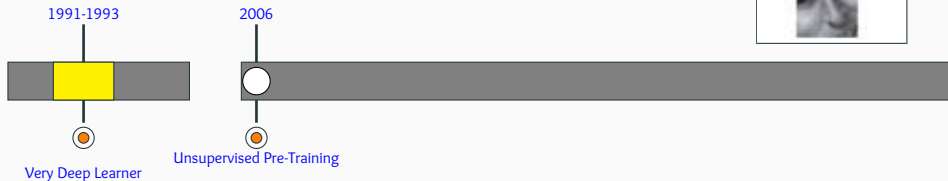
2006



Unsupervised Pre-Training

Unsupervised Pre-Training

The idea of unsupervised pre-training actually dates back to 1991-1993 (J. Schmidhuber) when it was used to train a “Very Deep Learner”



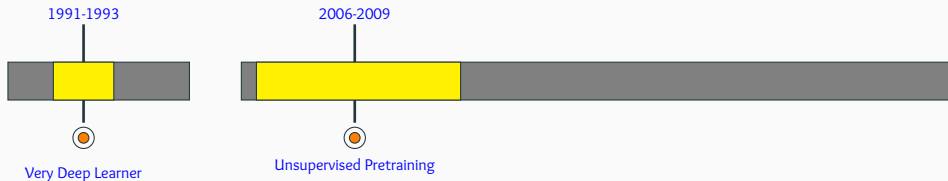
More insights (2007-2009)

Further Investigations into the effectiveness of Unsupervised Pre-training

Greedy Layer-Wise Training of Deep Networks

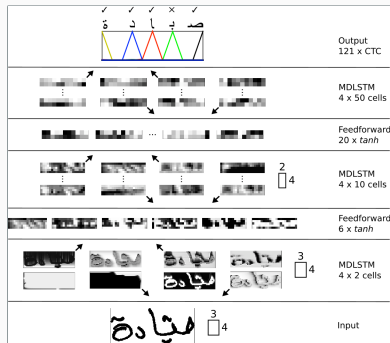
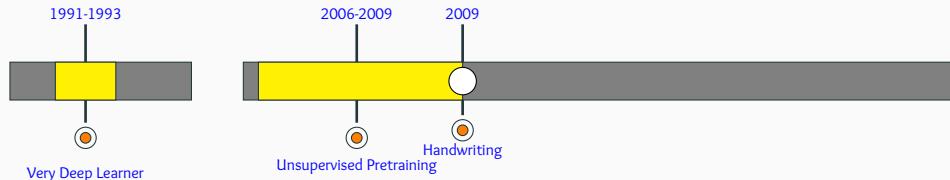
Why Does Unsupervised Pre-training Help Deep Learning?

Exploring Strategies for Training Deep Neural Networks



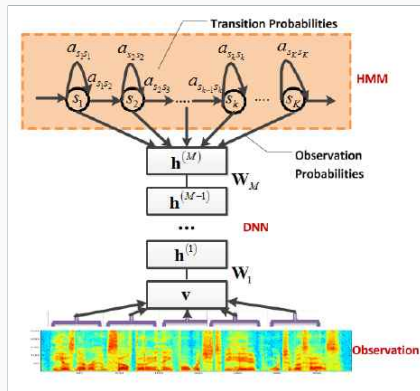
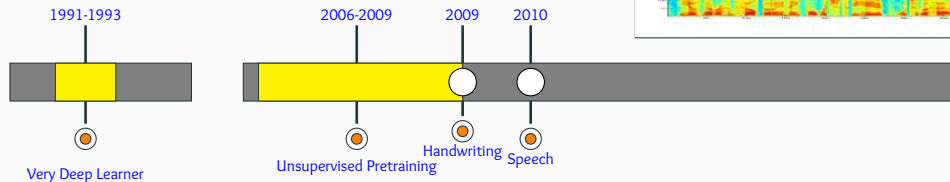
Success in Handwriting Recognition

Graves et. al. outperformed all entries in an international Arabic handwriting recognition competition^[9]



Success in Speech Recognition

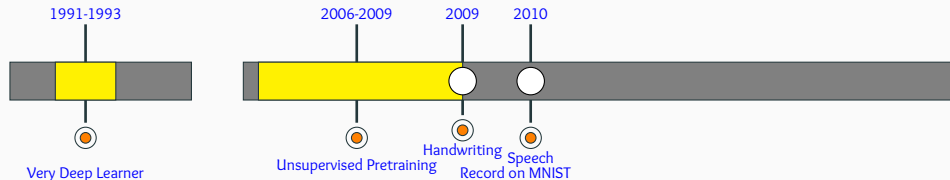
Dahl et. al. showed relative error reduction of 16.0% and 23.2% over a state of the art system^[10]



New record on MNIST

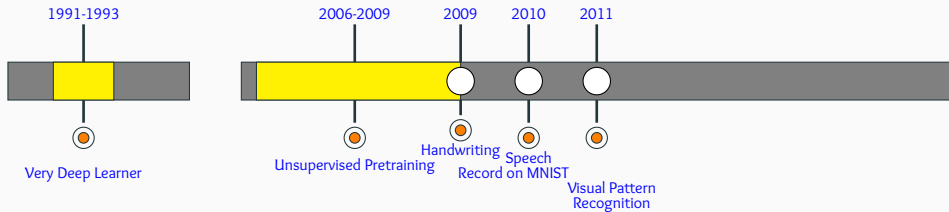
Ciresan et. al. set a new record on the MNIST dataset using good old backpropagation on GPUs (GPUs enter the scene) ^[11]

2 ² 17	1 ¹ 71	9 ⁸ 98	9 ⁹ 59	9 ⁹ 79	5 ⁵ 35	8 ⁸ 23
4 ⁹ 49	5 ⁵ 35	9 ⁴ 97	4 ⁹ 49	4 ⁴ 94	2 ² 02	5 ⁵ 35
6 ⁶ 16	4 ⁴ 94	0 ⁰ 60	6 ⁶ 06	6 ⁶ 86	1 ¹ 79	1 ¹ 71
9 ⁹ 49	0 ⁰ 50	5 ⁵ 35	8 ⁸ 98	9 ⁹ 79	7 ⁷ 17	1 ¹ 61
2 ⁷ 27	8 ⁸ 58	2 ² 78	6 ⁶ 16	6 ⁵ 65	4 ⁴ 94	6 ⁰ 60



First Superhuman Visual Pattern Recognition

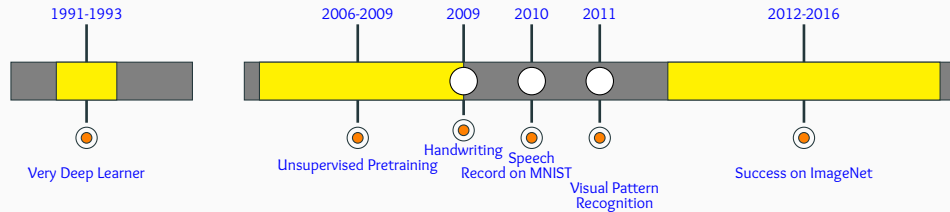
D. C. Ciresan et. al. achieved 0.56% error rate in the IJCNN Traffic Sign Recognition Competition^[12]



Winning more visual recognition challenges



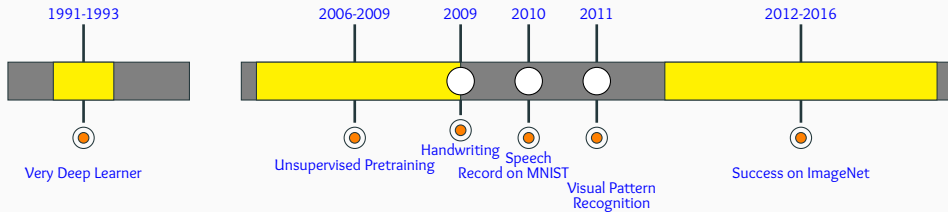
Network	Error	Layers
AlexNet ^[13]	16.0%	8



Winning more visual recognition challenges



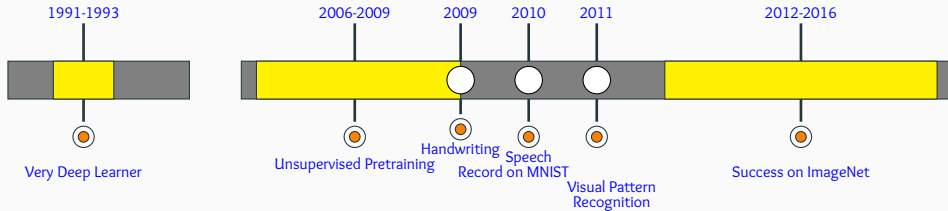
Network	Error	Layers
AlexNet ^[13]	16.0%	8
ZFNet ^[14]	11.2%	8



Winning more visual recognition challenges



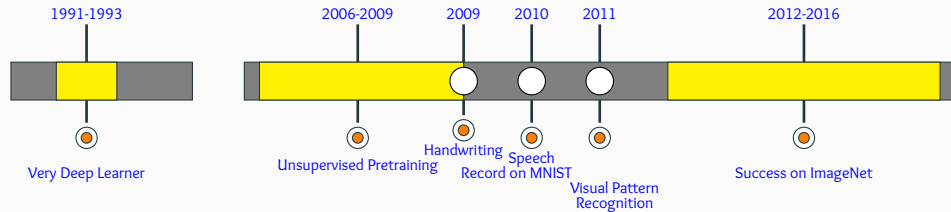
Network	Error	Layers
AlexNet ^[13]	16.0%	8
ZFNet ^[14]	11.2%	8
VGGNet ^[15]	7.3%	19



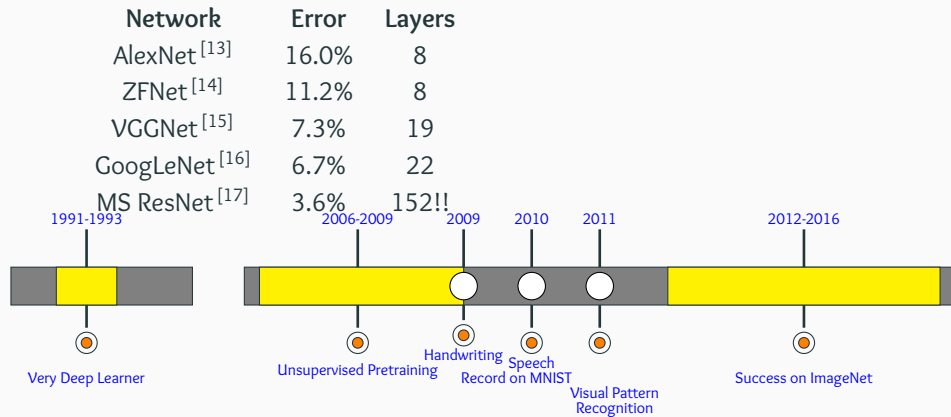
Winning more visual recognition challenges



Network	Error	Layers
AlexNet ^[13]	16.0%	8
ZFNet ^[14]	11.2%	8
VGGNet ^[15]	7.3%	19
GoogLeNet ^[16]	6.7%	22



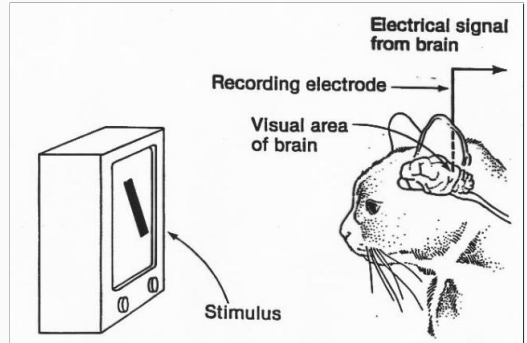
Winning more visual recognition challenges



Chapter 4: From Cats to Convolutional Neural Networks

Hubel and Wiesel Experiment

Experimentally showed that each neuron has a fixed receptive field - i.e. a neuron will fire only in response to a visual stimuli in a specific region in the visual space^[18]



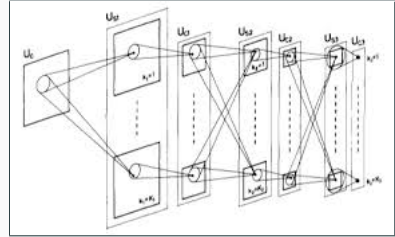
1959



H and W experiment

Neocognitron

Used for Handwritten character recognition and pattern recognition (Fukushima et. al.)^[19]



1959



H and W experiment

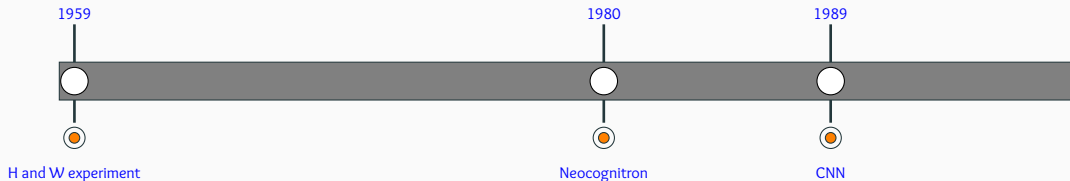
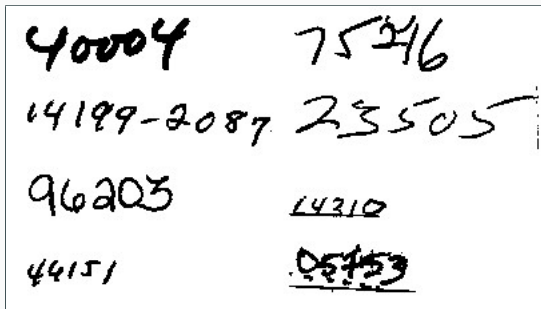
1980



Neocognitron

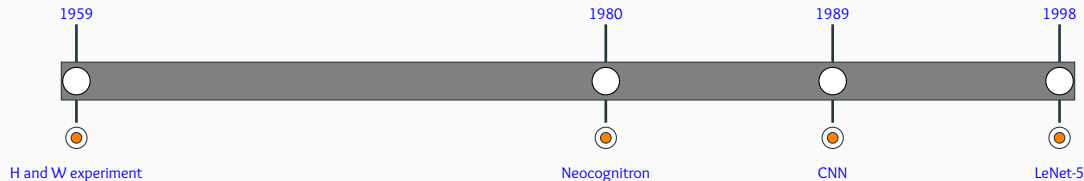
Convolutional Neural Network

Handwriting digit recognition using
backpropagation over a Convolutional Neural
Network (LeCun et. al.)^[20]



LeNet-5

Introduced the (now famous) MNIST dataset
(LeCun et. al.)^[21]

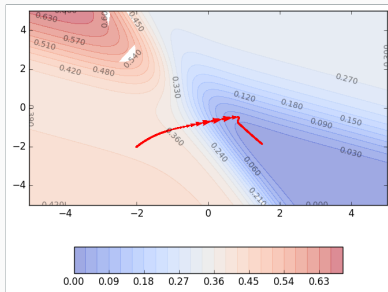


An algorithm inspired by an experiment on cats is today used to detect cats in videos :-)

Chapter 5: Faster, higher, stronger

Better Optimization Methods

Faster convergence, better accuracies



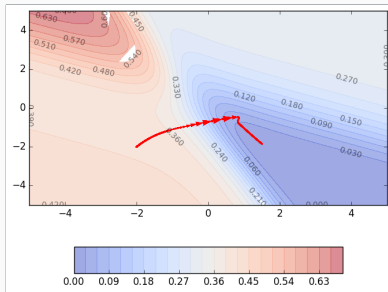
1983



Nesterov

Better Optimization Methods

Faster convergence, better accuracies



1983



Nesterov

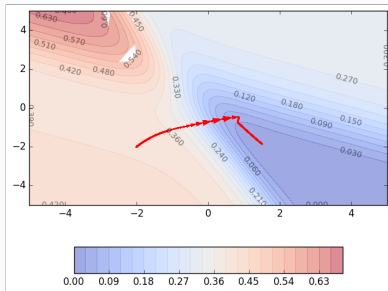
2011



Adagrad

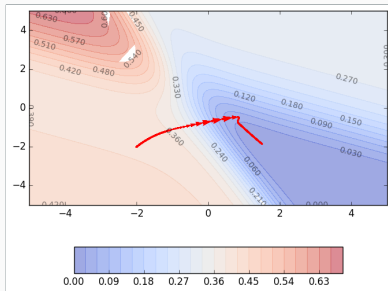
Better Optimization Methods

Faster convergence, better accuracies



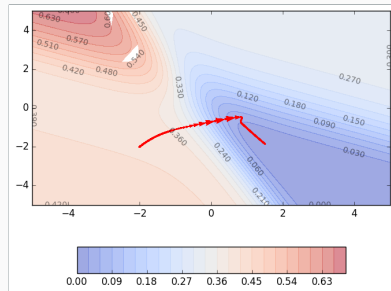
Better Optimization Methods

Faster convergence, better accuracies



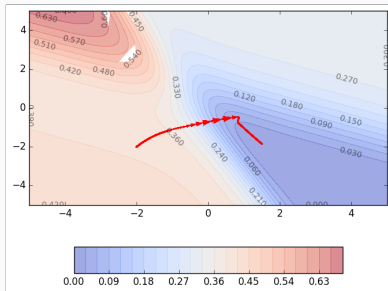
Better Optimization Methods

Faster convergence, better accuracies



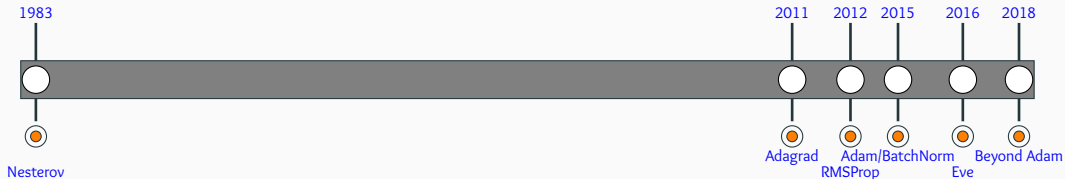
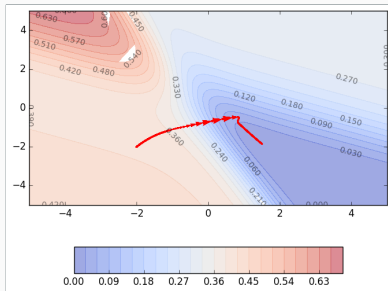
Better Optimization Methods

Faster convergence, better accuracies



Better Optimization Methods

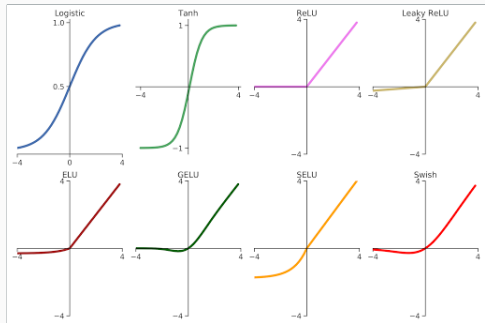
Faster convergence, better accuracies



Better Activation Functions

We have come a long way from the initial days when the logistic function was the default activation function in NNs!

Over the past few years many new functions have been proposed leading to better convergence and/or performance!



Chapter 6: The Curious Case of Sequences

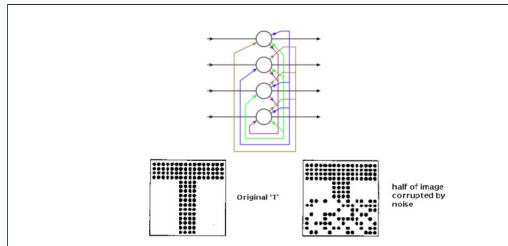
Sequences

- They are everywhere
- Time series, speech, music, text, video
- Each unit in the sequence interacts with other units
- Need models to capture this interaction



Hopfield Network

Content-addressable memory systems for storing and retrieving patterns^[22]



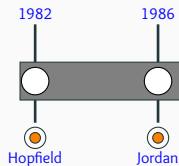
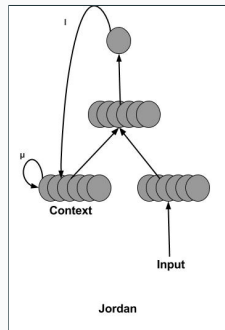
1982



Hopfield

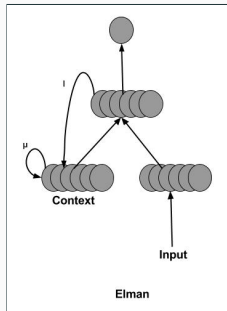
Jordan Network

The output state of each time step is fed to the next time step thereby allowing interactions between time steps in the sequence



Elman Network

The hidden state of each time step is fed to the next time step thereby allowing interactions between time steps in the sequence



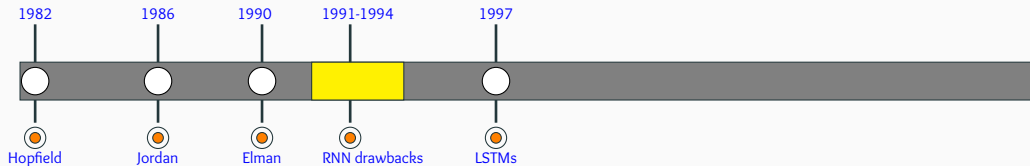
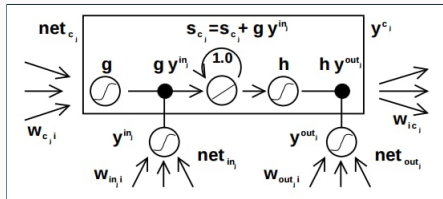
Drawbacks of RNNs

Hochreiter et. al. and Bengio et. al. showed the difficulty in training RNNs (the problem of exploding and vanishing gradients)



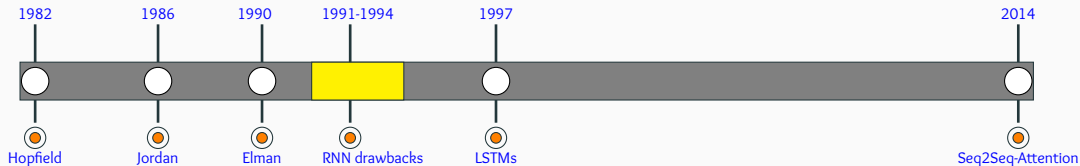
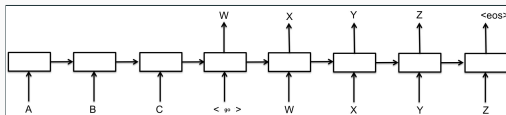
Long Short Term Memory

Showed that LSTMs can solve complex long time lag tasks that could never be solved before



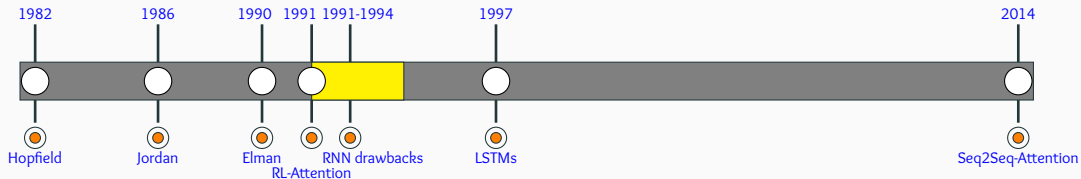
Sequence To Sequence Models

- Initial success in using RNNs/LSTMs for large scale Sequence To Sequence Learning Problems
- Introduction of Attention which is perhaps the idea of the decade!



RL for Attention

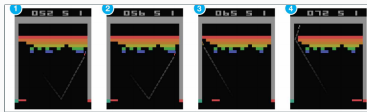
Schmidhuber & Huber proposed RNNs that use reinforcement learning to decide where to look



Chapter 7: Beating humans at their own game (literally)

Playing Atari Games

- Human-level control through deep reinforcement learning for playing Atari Games^[23]



2015



DQNs

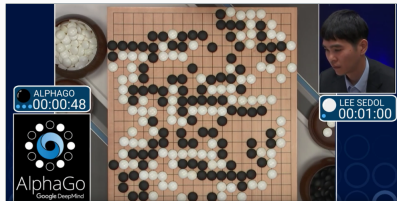
Let's GO

- Alpha Go Zero - Best Go player ever, surpassing human players^[24]
- GO is more complex than chess because of number of possible moves
- No brute force backtracking unlike previous chess agents

2015



DQNs
AlphaGO



Taking a shot at Poker

DeepStack defeated 11 professional poker players with only one outside the margin of statistical significance^[25]



2015

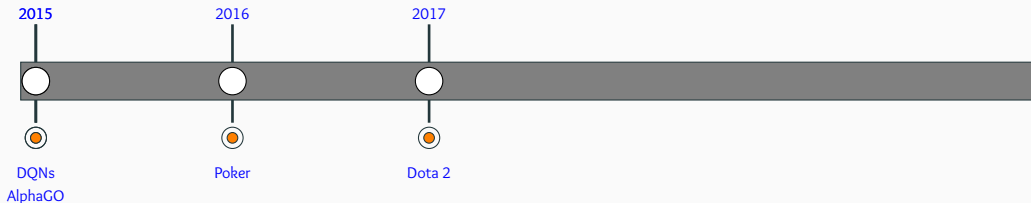
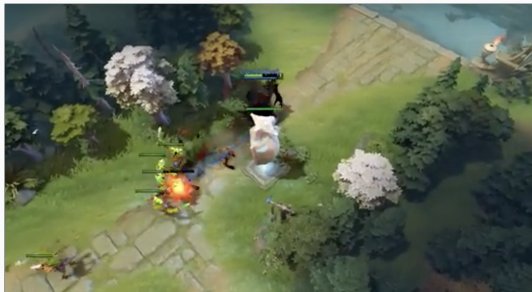
DQNs
AlphaGO

2016

Poker

Defense of the Ancients

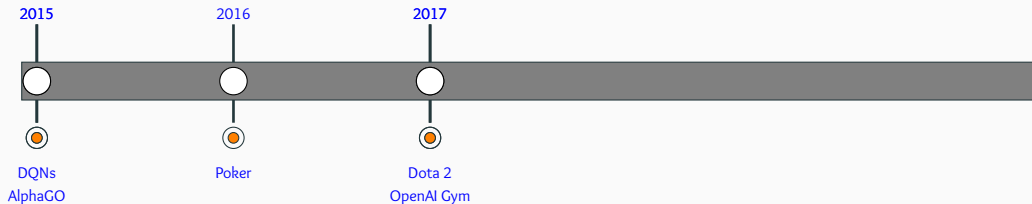
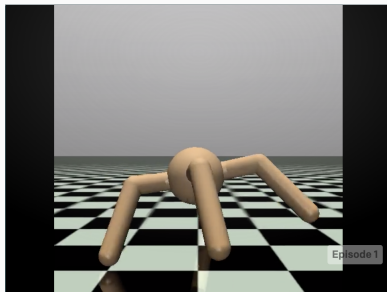
“Our Dota 2 AI, called OpenAI Five, learned by playing over 10,000 years of games against itself. It demonstrated the ability to achieve expert-level performance, learn human–AI cooperation, and operate at internet scale.” – *OpenAI*



A toolkit for RL

OpenAI Gym^a is a toolkit for developing and comparing reinforcement learning algorithms. It supports teaching agents everything from walking to playing games like Pong or Pinball.

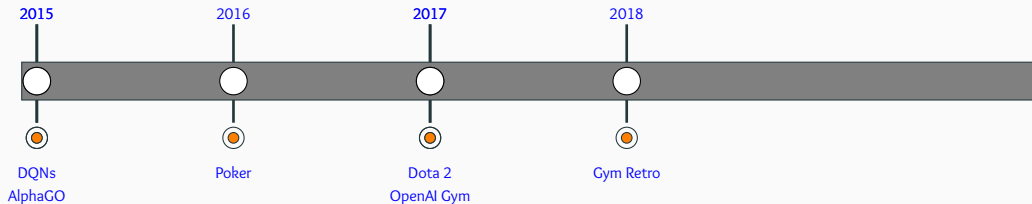
^a<https://gym.openai.com/>



RL for a 1000 games!

Open AI Gym Retro^a: a platform for reinforcement learning research on games which contains 1,000 games across a variety of backing emulators.

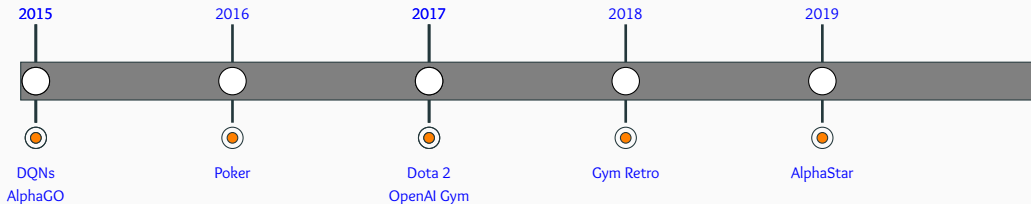
^a<https://openai.com/blog/gym-retro/>



Complex Strategy Games

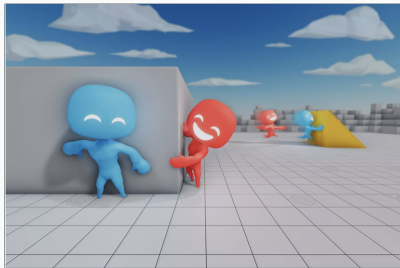
AlphaStar^a learned to balance short and long-term goals and adapt to unexpected situations while playing using the same maps and conditions as humans

^a<https://deepmind.com/>

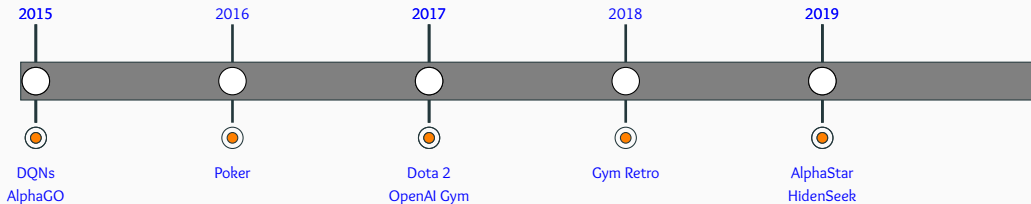


Learning to Hide

OpenAI demonstrated agents which can learn complex strategies such as chase and hide, build a defensive shelter, break a shelter, use a ramp to search inside a shelter and so on!

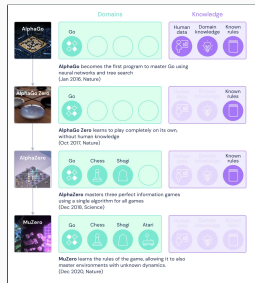


<https://openai.com/blog/emergent-tool-use/>

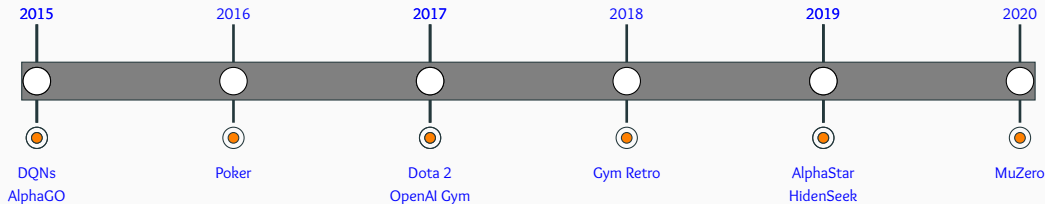


Jack of all, Master of all!

MuZero masters Go, chess, shogi and Atari without needing to be told the rules, thanks to its ability to plan winning strategies in unknown environments.



<https://deepmind.com/blog>



Chapter 8: The Madness (2013-)

He sat on a chair.

Language Modeling

Mikolov et al. (2010) ^[26]

Kiros et al. (2015) ^[27]

Kim et al. (2015) ^[28]



Speech Recognition

Hinton et al. (2012) ^[29]

Graves et al. (2013) ^[30]

Chorowski et al. (2015) ^[31]

Sak et al. (2015) ^[32]

MACHINE TRANSLATION



Machine Translation

Kalchbrenner et al. (2013) ^[33]

Cho et al. (2014) ^[34]

Bahdanau et al. (2015) ^[35]

Jean et al. (2015) ^[36]

Gulcehre et al. (2015) ^[37]

Sutskever et al. (2014) ^[38]

Luong et al. (2015) ^[39]

Zheng et al. (2017) ^[40]

Cheng et al. (2016) ^[41]

Chen et al. (2017) ^[42]

Firat et al. (2016) ^[43]